



भारतीय राष्ट्रीय मानक संस्थान
NATIONAL STANDARDS BODY OF INDIA



भारत सरकार
GOVERNMENT OF INDIA

भारतीय मानक ब्यूरो
BUREAU OF INDIAN STANDARDS
उपभोक्ता मामले, खाद्य एवं सार्वजनिक वितरण मंत्रालय
MINISTRY OF CONSUMER AFFAIRS,
FOOD & PUBLIC DISTRIBUTION
9, Bahadur Shah Zafar Marg, New Delhi 110002

व्यापक परिचालन मसौदा

हमारा संदर्भ : सीईडी 13/टी-45

21 अगस्त 2026

तकनीकी समिति : भवन निर्माण की रीतियाँ/विषय समिति, सीईडी 13

प्राप्तकर्ता :

- क) सिविल अभियांत्रिकी विभाग परिषद, सीईडीसी के सभी सदस्य
- ख) सीईडी 13 के सभी सदस्य
- ग) रुचि रखने वाले अन्य निकाय।

प्रिय महोदय/महोदया,

निम्नलिखित भारतीय मानक का मसौदा संलग्न है:

प्रलेख संख्या	शीर्षक
सीईडी 13 (26762) WC	नाइट्रोसेलुलोज एवं कोल्ड कैटेलाइज्ड सामग्री द्वारा लकड़ी एवं लकड़ी-आधारित उत्पादों की फिनिशिंग — रीति संहिता का भारतीय मानक मसौदा (IS 4597 का पहला पुनरीक्षण) (ICS 79.020)

कृपया इस मसौदे का अवलोकन करें और अपनी सम्मतियाँ यह बताते हुए भेजे कि यह मसौदा प्रकाशित हो तो इन पर अमल करने में आपको व्यवसाय अथवा कारोबार में क्या कठिनाइयाँ आ सकती हैं।

सम्मतियाँ भेजने की अंतिम तिथि : **07 अक्टूबर 2026**

सम्मति यदि कोई हो तो कृपया अधोहस्ताक्षरी को ई-मेल द्वारा nakkalapraiveen.sai@bis.gov.in पर या उपरलिखित पते पर, संलग्न फॉर्मेट में भेजें। सम्मतियाँ बीआईएस ई-गवर्नेंस पोर्टल, www.manakonline.in के माध्यम से ऑनलाइन भी भेजी जा सकती हैं।

यदि कोई सम्मति प्राप्त नहीं होती है अथवा सम्मति में केवल भाषा संबंधी त्रुटि हुई तो उपरोक्त प्रलेख को यथावत अंतिम रूप दे दिया जाएगा। यदि सम्मति तकनीकी प्रकृति की हुई तो विषय समिति के अध्यक्ष के परामर्श से अथवा उनकी इच्छा पर आगे की कार्यवाही के लिए विषय समिति को भेजे जाने के बाद प्रलेख को अंतिम रूप दे दिया जाएगा।

यह प्रलेख भारतीय मानक ब्यूरो की वेबसाइट www.bis.gov.in पर भी उपलब्ध है।

धन्यवाद।

भवदीय,

द्वैपायन भद्र
प्रमुख (सिविल इंजीनियरिंग)

संलग्न: उपरलिखित



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WIDE CIRCULATION DRAFT

Our Ref.: CED 13/T-45

21 August 2026

Technical Committee: Building Construction Practices Sectional Committee, CED 13

ADDRESSED TO:

- a) All Members of Civil Engineering Division Council, CEDC
- b) All Members of CED 13
- c) All others interested.

Dear Sir/Madam,

Please find enclosed the following draft:

Doc. No.	Title
CED 13 (26762) WC	Draft Indian Standard Finishing of Wood and Wood Based Products with Nitrocellulose and Cold Catalysed Materials — Code of Practice (First Revision of IS 4597) (ICS 79.020)

Kindly examine the attached draft and forward your views stating any difficulties which you are likely to experience in your business or profession, if this is finally adopted as National Standard.

Last Date for Comments: **07 October 2026**

Comments if any, may please be made in the enclosed format and emailed at nakkalapraveen.sai@bis.gov.in or sent at the above address. Additionally, comments may be sent online through the BIS e-governance portal, www.manakonline.in.

In case no comments are received or comments received are of editorial nature, kindly permit us to presume your approval for the above document as finalized. However, in case comments, technical in nature are received, then it may be finalized either in consultation with the Chairman, Sectional Committee or referred to the Sectional Committee for further necessary action if so desired by the Chairman, Sectional Committee.

The document is also hosted on BIS website www.bis.gov.in.

Thanking you,

Yours faithfully,

Dwaipayan Bhadra
Head (Civil Engineering)

Encl: As above

FORMAT FOR SENDING COMMENTS ON BIS DOCUMENTS

(Please use A-4 size sheet of paper only and type within fields indicated. Comments on each clause/sub-clause/table/fig etc. be started on a fresh box. Information in column 3 should include reasons for the comments and suggestions for modified working of the clauses when the existing text is found not acceptable. Adherence to this format facilitates Secretariat's work) {Please e-mail your comments to nakkalapraveen.sai@bis.gov.in}

Doc. No.: CED 13 (26762) WC

Title: Draft Indian Standard Finishing of Wood and Wood Based Products with Nitrocellulose and Cold Catalysed Materials — Code of Practice
(First Revision of IS 4597) (ICS 79.020)

LAST DATE OF COMMENTS: **07/10/2026**

NAME OF THE COMMENTATOR/ORGANIZATION: _____

Sl. No.	Clause/Sub-clause/Para No.	Comments/Suggestions	Modified Wordings	Reasons/Justifications for the Proposed Changes
(1)	(2)	(3)	(4)	(5)

NOTE – Kindly insert more rows as necessary for each clause/table, etc

BUREAU OF INDIAN STANDARDS

DRAFT FOR COMMENTS ONLY

(Not to be reproduced without the permission of BIS or used as an Indian Standard)

Draft Indian Standard

**FINISHING OF WOOD AND WOOD BASED PRODUCTS WITH NITROCELLULOSE AND
COLD CATALYSED MATERIALS — CODE OF PRACTICE**

(First Revision of IS 4597)

ICS 79.020

Building Construction Practices
Sectional Committee CED 13

Last date for comment:
07 October 2026

FOREWORD

(Formal clauses will be added later)

The term wood finishing covers all the operations involved in the application of transparent and semi-transparent materials to wood surfaces to enhance the natural beauty of wood and to maintain the desired surface appearance. The natural colour of the wood may or may not be altered in the process.

The other equally important purpose of wood finishing is to protect the surface of wood from atmospheric conditions which would otherwise cause the wood to rot or change its dimensions. It also protects the wood from spotting and discolouration caused by dust, gases, grease or handling.

This standard was first published in 1968. This revision has been brought out to bring the standard in the latest style and format of the Indian Standards.

In the formulation of this standard, due weightage has been given to international coordination among the standards and practices prevailing in different countries in addition to relating it to the practices in the field in this country.

For the purpose of deciding whether a particular requirement of this standard is complied with, the final value, observed or calculated, expressing the result of a test or analysis, shall be rounded off in accordance with IS 2 : 2022 ‘Rules for rounding off numerical values (*second revision*)’. The number of significant places retained in the rounded-off value should be the same as that of the specified value in this standard.

*Draft Indian Standard***FINISHING OF WOOD AND WOOD BASED PRODUCTS WITH
NITROCELLULOSE AND COLD CATALYSED MATERIALS —
CODE OF PRACTICE***(First Revision)***1 SCOPE**

This standard deals with the finishing of wood and wood based products using nitrocellulose-based materials and cold-catalysed materials.

2 REFERENCES

The standards given below contain provisions which through reference in this text, constitute provisions of this standard. At the time of publication, the editions indicated were valid. All standards are subject to revision and parties to agreements based on this standard are encouraged to investigate the possibility of applying the most recent edition of these standards:

<i>IS No.</i>	<i>Title</i>
IS 287: 1993	Permissible moisture content for timber used for different purposes — Recommendations (<i>third revision</i>)
IS 707 : 2011	Timber technology and utilization of wood, bamboo and cane — Glossary of terms (<i>third revision</i>)
IS 1303 : 2024	Paints — Glossary of terms (<i>third revision</i>)

3 TERMINOLOGY

For the purpose of this standard, the definitions of terms relating to finishing shall be as given in IS 1303 and those relating to wood and related material shall be as given in IS 707.

4 NECESSARY INFORMATION

4.1 The decorative and protective value of a finish may be influenced by the nature of the surface on which it is applied. There are several species of wood and an increasingly large number of wood-based panel products often differing from each other in surface characteristics. They may be hard, soft, resinous or porous in varying degrees.

4.1.1 Wood is a hygroscopic material and tries to reach a state of equilibrium with the atmosphere in so far as its moisture content is concerned. Changes in moisture content are accompanied by swelling and shrinkage, which is most pronounced across the grain of the wood. Due to moisture movement the summerwoods swell to a greater extent than springwoods and this sets up stress-concentration at the sharply defined junctions between one year's summer growth and the next year's spring and the failure commences about this region. The resulting stresses on the film of finishing material are such as to cause fissures to develop along the grain under adverse circumstances.

4.1.2 Before painting, wood shall be properly seasoned and the moisture content shall be brought as near as possible to the equilibrium moisture content as given in IS 287 so as to prevent uneven shrinkage

during drying which may result in distortion or even in cracks in the finish. It is also advisable not to finish excessively dry wood.

4.1.3 The cellular structure of wood has a strong influence on the absorption of liquid components of finishes. Non-uniform absorption of the vehicle by the wood upsets the pigment-binder ratio, impairing the appearance and life of the coating.

4.1.4 Plywood, blockboard, hardboard and particle board have greater dimensional stability than solid wood and provide more uniform surface for finishing.

4.2 For efficient application of wood finishes the following information is necessary and shall be taken into consideration:

- a) The type of wood and nature of its pre-treatment, namely, preservation and seasoning;
- b) The location in which it is intended to be used;
- c) The atmospheric conditions in the locality namely, temperature and humidity; and
- d) The type of finish to be applied.

5 PREPARATION OF WOOD FOR FINISHING

5.1 Cleaning

Use a damp cloth with mild soap to clean the surface and remove any impurities.

5.2 Sanding

Adequate sanding of surfaces to be finished at all stages of finishing is a major factor in achieving a satisfactory result. Abrasions, dents, and deep sanding marks will show up in clear finishes. The surfaces shall be prepared by careful smoothing with 320 grit sandpaper used in the direction of the grain. Scratches across the grain, coarse sanding marks and bruises are likely to become stained darker than the rest of the surface thereby spoiling the finished appearance. If water stain is to be used, the surface shall be wetted with water to raise the grain and then allowed to dry before finally smoothing. Machine sanding or belt sanding is recommended.

5.3 The surfaces shall be carefully wiped free of wood flour and any minor repairs that may be necessary shall be made.

6 FILLING

6.1 The primary function of fillers is to fill the pores and opened cells of the wood in the surface layer. This is necessary to prevent the excessive penetration of the finish that is subsequently applied and to level off the surface of a porous wood to make a smooth top finish possible.

6.2 For special stain effects, coloured fillers are used. However, the result lacks the grain and colour contrast characteristic of wood stains.

6.3 The filler is applied by a rag rolled to form a pad. The application is in a circular direction with sufficient pressure to force the filler into the pores. It may also be sprayed and then a specially designed mechanical mop may be used to force the filler into the pores and remove the excess. The number of coats applied by either method depends on the porosity of the wood.

6.4 The excess filler should be wiped off with a dry rag within 5 min to 10 min of application.

6.5 The filler is air dried overnight and levelled off by dry sanding with 180 grit sandpaper.

7 SEALING

7.1 The primary function of the sealer is to prevent the swelling and lifting of the grains, bleeding of wood filler, etc, on application of the finish. However, in most cases, this is not essential in which case the sealer coat(s) may be replaced with finish coat(s).

7.2 The sealer is nitrocellulose based if the final finish is of the same material and is applied by spray. The number of coats required depends upon the non-volatile content of the material used.

7.3 For wood finish with cold catalysed material the sealer used may be single pack like shellac sealer or two pack like cold catalysed sealer. In both cases, the application is by spray.

7.4 The sealer is allowed to dry hard normally overnight, and sanded with 320 sandpaper paper taking care not to cut through the edges and corners. The dust is blown off and the surface wiped with a rag.

8 LEVELLING WITH NITROCELLULOSE WOOD FILLER

If the sealer shows uneven surface, necessitating levelling off at this stage, a transparent, nitrocellulose based wood filler is used. It is applied with a putty knife in straight strokes; air dried for one hour and dry sanded with 320 sandpaper.

9 STAINING

9.1 Staining of wood may be used for indoor fittings only when followed by clear finish. The object of staining wood is to darken it as part of a decorating scheme. If skilfully carried out, staining may be used with good effect to enhance the natural grain or figuring of the wood. The stains may be applied over wood filler or over the sealer.

9.2 The ability of stains to colour wood is affected by wood structure, the texture and grain of any given piece, the type of stain, its concentration, etc. Staining tends to accentuate the natural variations in grain. In hardwoods it results in increased emphasis on grain variations traceable to porous spring wood and dense summer wood. In soft woods on the contrary, the springwood generally absorbs more stain and hence the natural sequence of colour shades is reversed.

9.3 Depending on the requirements a water, spirit, naphtha or acid resistant stain may be used. It may be applied by brushing, sponging and wiping or spraying. The stain shall be so thinned that it can be applied fairly liberally without over-staining. Care shall be taken, especially on absorbent softwoods, to apply the stain evenly and without over-lapping. Spirit stains, in particular, require careful and quick application as they dry very quickly. In general, flat surfaces shall be treated first and mouldings and edges last, the object being to avoid double staining along the edges.

9.3.1 Water Stains

Water stains are made with water soluble dyes or water miscible pigment paste. They emphasize the grain, especially that of softwoods, since they are readily absorbed by the porous portions but less readily by the denser, more resinous portions. They will raise the grain of the wood, thus spoiling the smoothness of the finish if a highly polished effect is required; this difficulty can be overcome by first wetting the surface with water to raise the grain and then, after drying, smoothing it with 240 grit sandpaper before staining. Where it is necessary to provide a temporary staining treatment on wood, that is, unseasoned, water stain is preferable to other types of stain.

9.3.2 Spirit Stains

Spirit stains are solutions of spirit soluble dyes in industrial methylated spirit. Like water stains, spirit stains penetrate more into softer portions of the wood and so accentuate the grain but they do not cause the fibres to swell nor raise the grain. They will dry very quickly and shall be applied quickly and skilfully to avoid patchy effects. If applied to damp wood, the dyes in the stains are liable to be thrown out of solution. The surface after staining with spirit stains may be finished in the same way as after treating with water stains.

10 FINISHING

10.1 The function of the finish is to decorate and protect the wood. It is nitrocellulose based or a two-pack material (a lacquer and an activator) which has limited pot life after mixing and, therefore, only the amount required for use shall be mixed and used up within the recommended time.

10.2 It may be applied over the sealer or, in many cases, directly over the filler. The application is by spray and the number of coats depends upon the non-volatile content of the material used.

10.2.1 In the case of nitrocellulose type of finish, it is recommended that at a time two coats are sprayed wet on wet, with 15 min to 30 min air drying between coats; air dried overnight and wet flatted with 600 grit sandpaper and soap water before spraying the next two coats. In place of sealer the finish coats may be applied after checking that it has no effect like grain raising or bleeding of wood filler. In addition to the sealer coats two or three coats of finish are considered adequate for furniture whereas four to six coats are required for radio cabinets.

10.2.2 In the case of cold catalysed type of finish, in place of the sealer, a coat of finish may be applied after checking that it has no effect like grain raising or bleeding of wood filler. Each coat of finish is air dried for at least 4 h so as to be dry enough for subsequent operations. Before recoating it shall be ensured that the surface is clean and dry. In addition to the sealer coat, one coat of finish is considered adequate for furniture whereas two or three coats of finish are required for radio cabinets.

10.3 After the application of the final coat, the film is allowed to age for at least 24 h.

11 PULLING OVER

11.1 Pull over solution may be used to level off only the nitrocellulose finish. However, if burnishing and polishing is going to be done this operation may be eliminated.

11.2 The application is by pad. The pad should be moderately damp when the operation is started. The initial application is along the grain and then, as the pad dries, the pressure is increased and a circular motion is used. This works out all the marks in the film and the final straightening-up motion leaves a bright and full finish.

11.3 The pullover solution leaves some solvent in the finish and hence at least 12 h air drying is required before handling and packing. However, the fittings may be fixed during this period.

11.4 For best results pulling over is followed up with a polish.

12 BURNISHING

12.1 The best appearance may be obtained from nitrocellulose and cold-catalysed type of finishes by means of burnishing.

12.2 Levelling before burnishing is advisable. The film is wet sanded with a 600 grit sandpaper and soap water. The sanding is done along the grain, taking care not to cut through the edges and corners.

12.3 A suitable burnishing compound is used for further levelling. The application is by pad.

12.4 A suitable polishing compound is used to clean the surface after burnishing and to bring up the gloss. The application is done along the grain with a pad.

12.5 A final light rub with a flexible shaft polishing machine using a soft lamb's wool mop gives the best results. Small items which can be easily manipulated may be mopped on a stationary polishing machine. Care shall be taken to avoid burning the finish by local overheating.

12.6 In case of semi glossy or matt finishes with nitrocellulose based materials burnishing is not required. However, wet sanding with 600 grit sandpaper and soap water or burnishing may be done to obtain a smooth even surface.